

eCHT research seminar
September 22, 2026

PRIME IDEALS IN THE BURNSIDE TAMBARA FUNCTOR

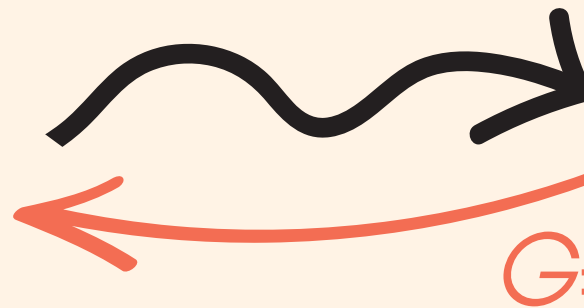
MAXINE E. CALLE
UNIVERSITY OF BONN

callem@math.uni-bonn.de

jt. with S. Ginnett [2020] and D. Chan, D. Mehrle,
J.D. Quigley, B. Spitz, and D. Van Niel [2025]

OUTLINE

non-equivariant story



equivariant story

NON-EQUIVARIANT THEOREM (?)

$$\mathrm{Spec}(\mathbb{Z}) = \{ (p) \mid p \text{ prime or } 0 \}$$

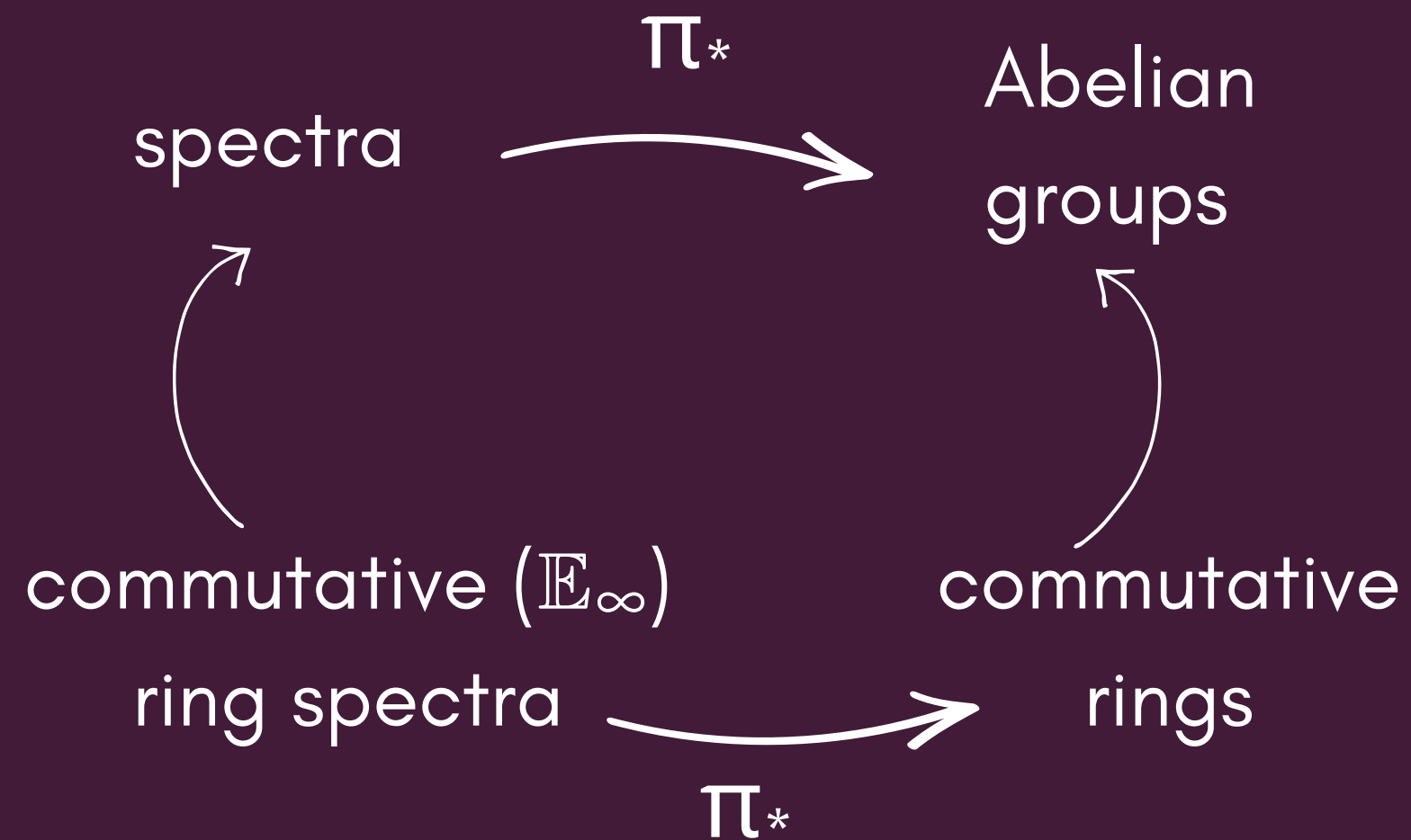
- what??
- why?
- how?

EQUIVARIANT THEOREM (Nakaoka, C.-Ginnett, C.-CMQSVN)

$$\mathrm{Spec}(\underline{A}_G) = \{ P_{H,p} \mid p \text{ prime or } 0, H \leq G \}$$

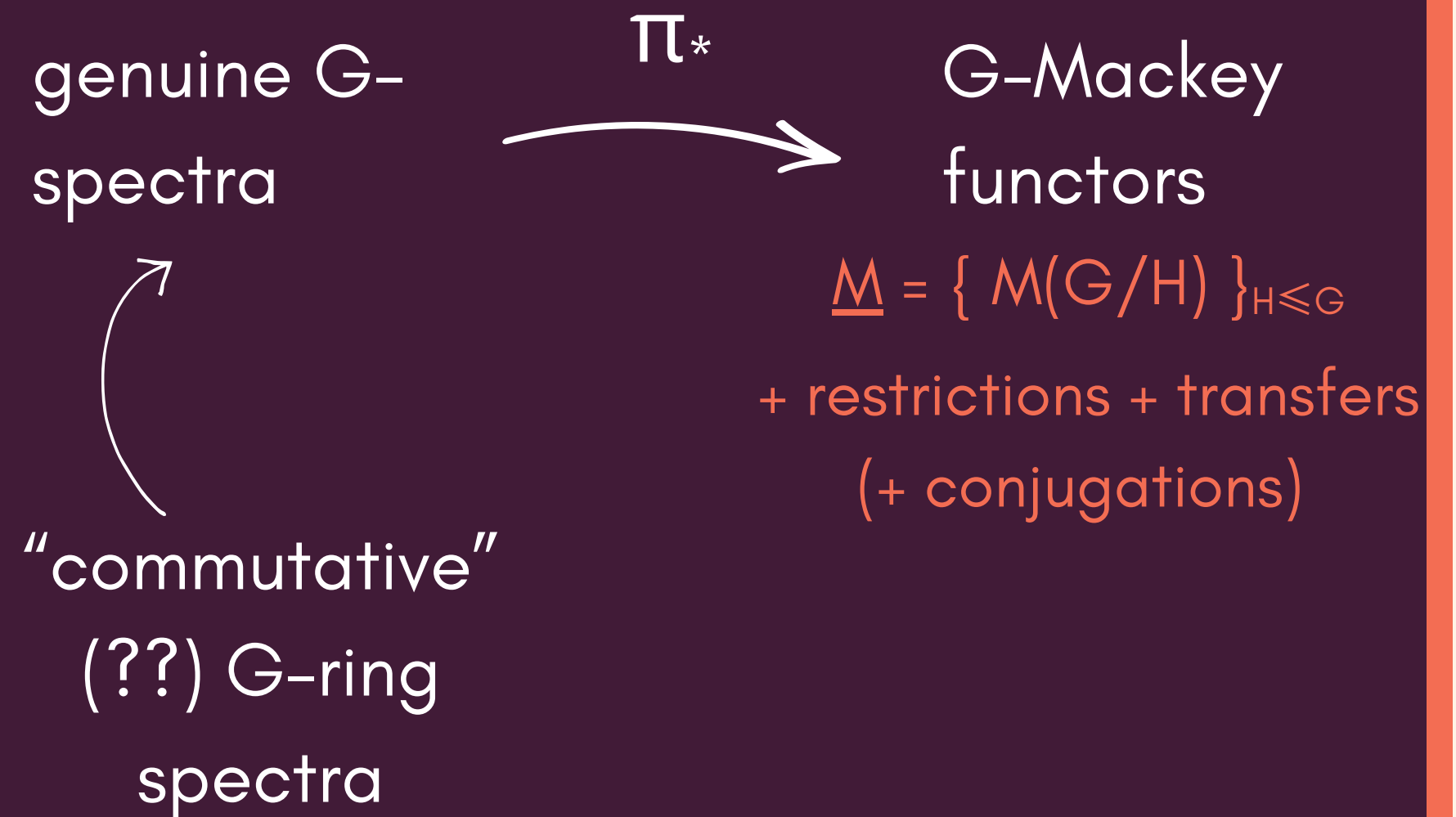
WHAT IS THE BURNSIDE TAMBARA FUNCTOR?

STABLE HOMOTOPY THEORY



EX. $\pi_0(\mathbb{S}) \cong \mathbb{Z}$

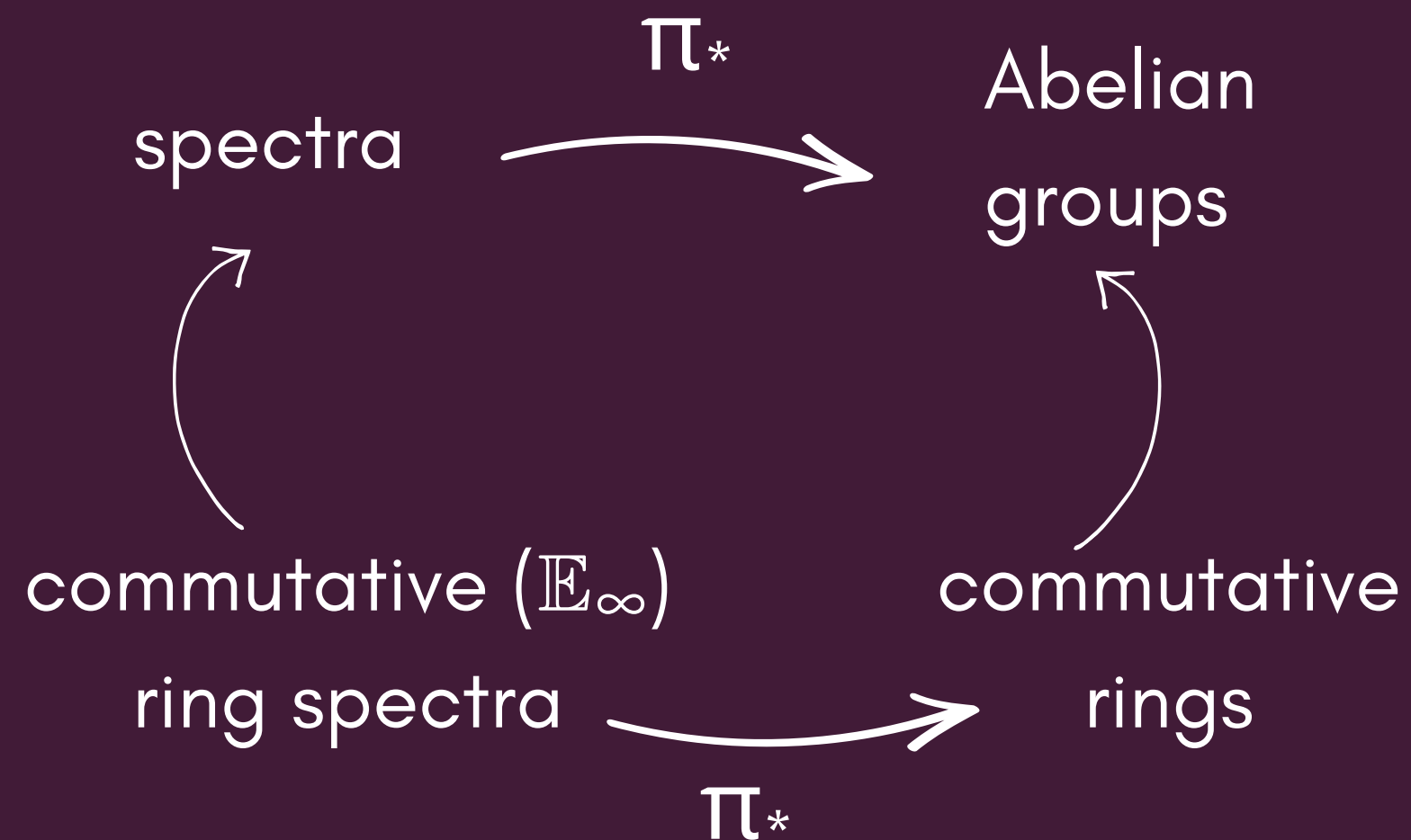
STABLE EQUIVARIANT HOMOTOPY THEORY



EX. $\pi_0(\mathbb{S}_G) \cong \underline{A}_G$

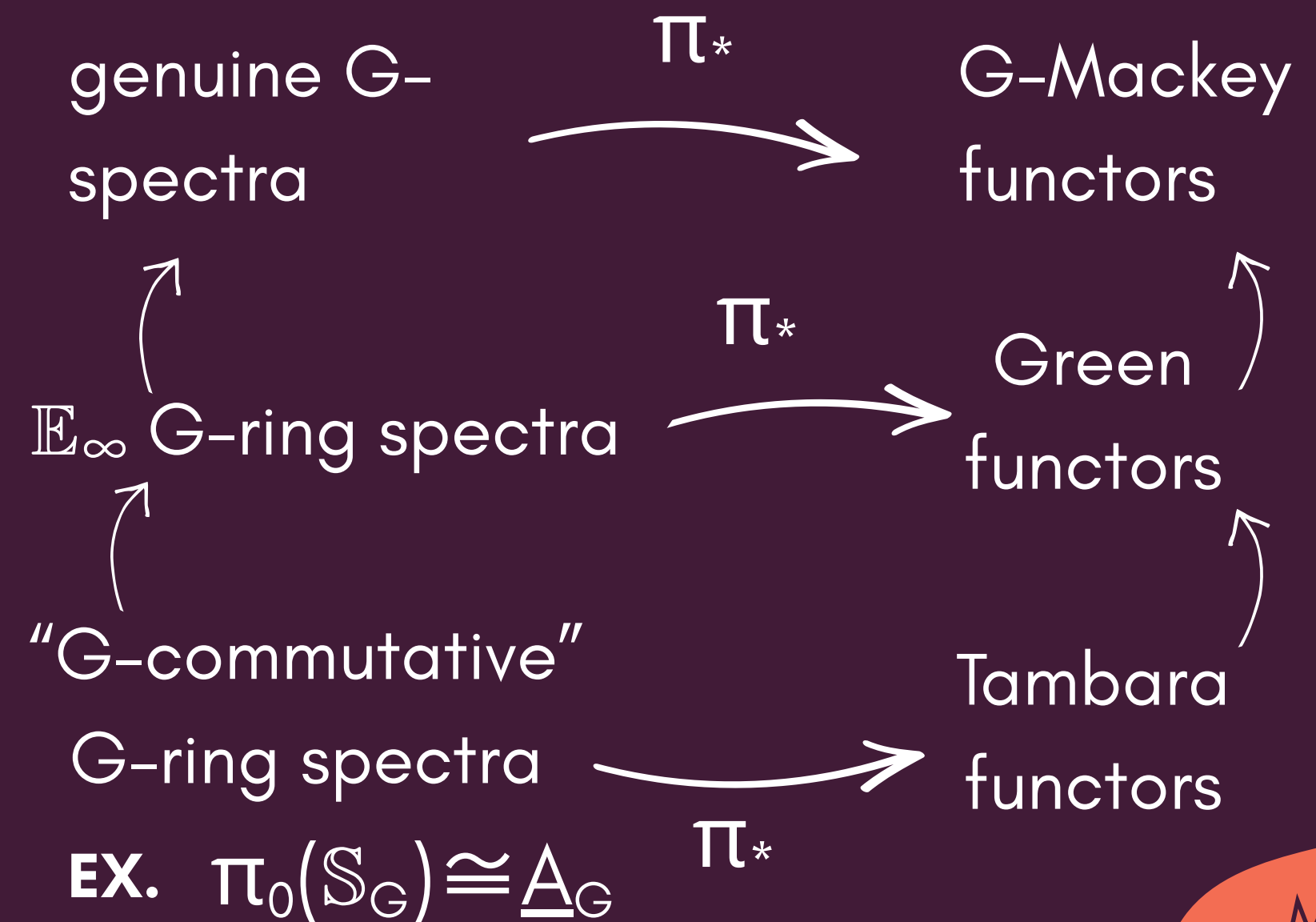
WHAT IS THE BURNSIDE TAMBARA FUNCTOR?

STABLE HOMOTOPY THEORY



EX. $\pi_0(S) \cong \mathbb{Z}$

STABLE EQUIVARIANT HOMOTOPY THEORY



EX. $\pi_0(S_G) \cong \underline{A}_G$

Mackey
functor +
norms

WHAT IS THE BURNSIDE TAMBARA FUNCTOR?

It's the π_0 -
Tambara functor
of S_G !

:(

EX. $G = C_q$, q prime

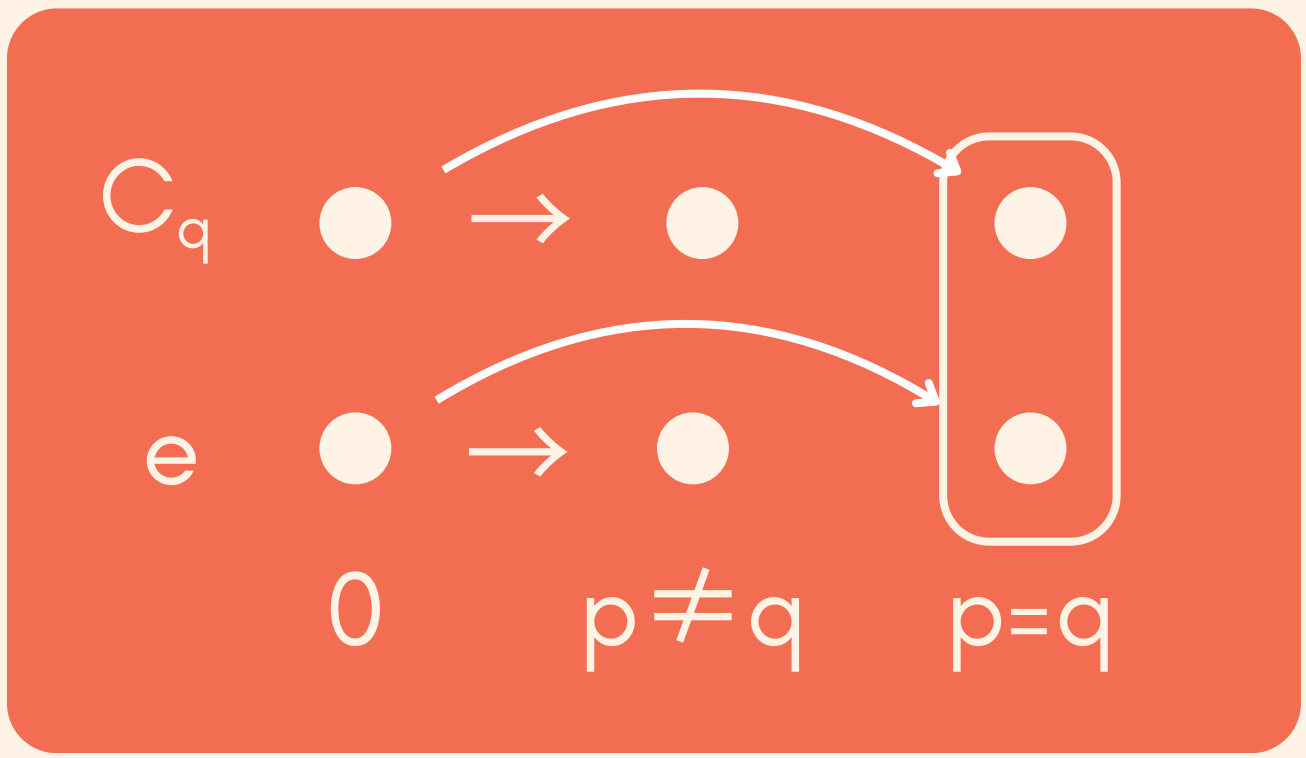
$$A(C_q) \cong \mathbb{Z}[x]/(x^2 - qx)$$

$$C_q/C_q \rightarrow 1$$

$$C_q/e \rightarrow x$$

$$\begin{aligned} \ker(\phi_{e,p}) &= \{ X \mid |X| \equiv_p 0 \} \\ &= \langle x - q, p \rangle \end{aligned}$$

$$\begin{aligned} \ker(\phi_{G,p}) &= \{ X \mid |X^G| \equiv_p 0 \} \\ &= \langle x, p \rangle \end{aligned}$$



WHAT IS $\underline{A}_G$ (REALLY)

DEF. The Burnside ring of G is $A(G) = \text{Gr}(\text{Fin}_G, \sqcup, \times) \cong \mathbb{Z}[\{G/H\}_{H \leq G}]/\sim$

THM. (Dress, 1969)

$$\text{Spec}(A(G)) = \{ \ker(\phi_{H,p}) \mid p \text{ prime or } 0, H \leq G \}$$

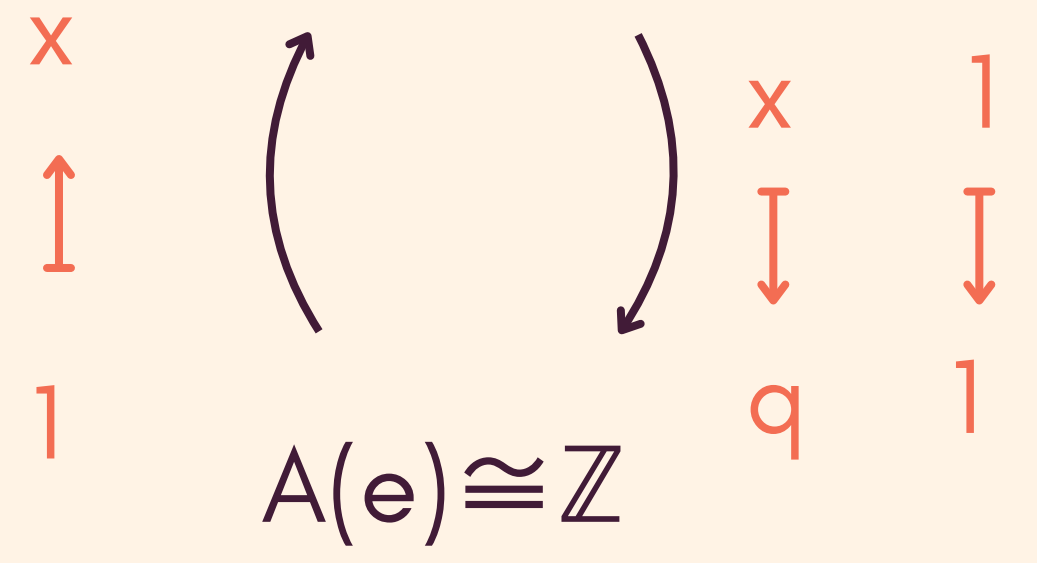
where $\phi_{H,p}: A(G) \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$

$$X \rightarrow |X^H| \bmod p$$

PF IDEA: Study “ghost map” $\phi_G: A(G) \rightarrow \prod_{H \leq G} \mathbb{Z}$ and the induced map on Spec

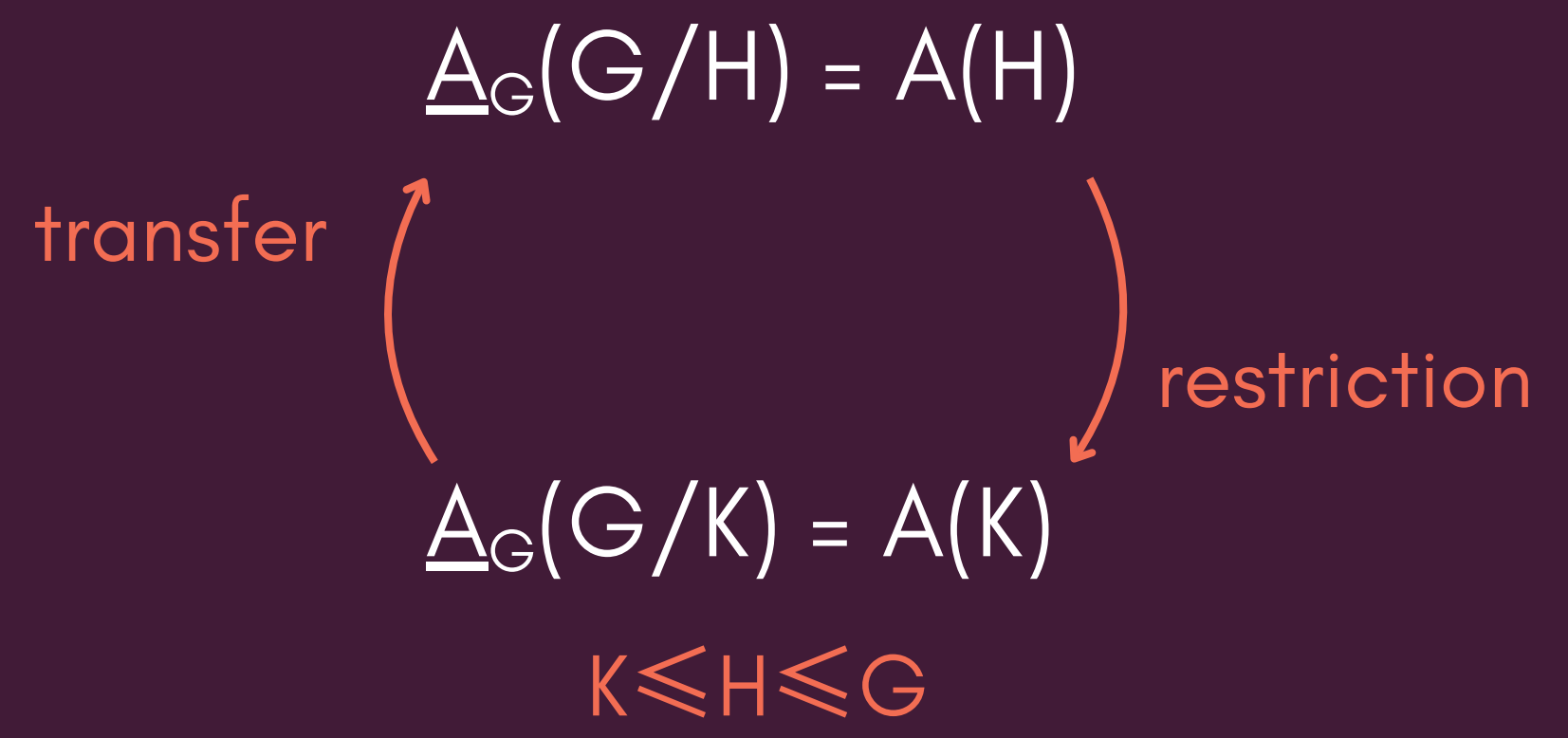
EX. $G = C_q$, q prime

$$A(C_q) \cong \mathbb{Z}[x]/(x^2 - qx)$$



WHAT IS $\underline{A}_G$ (REALLY)

“DEF.” The Burnside Mackey functor of G “is” $\underline{A}_G$:

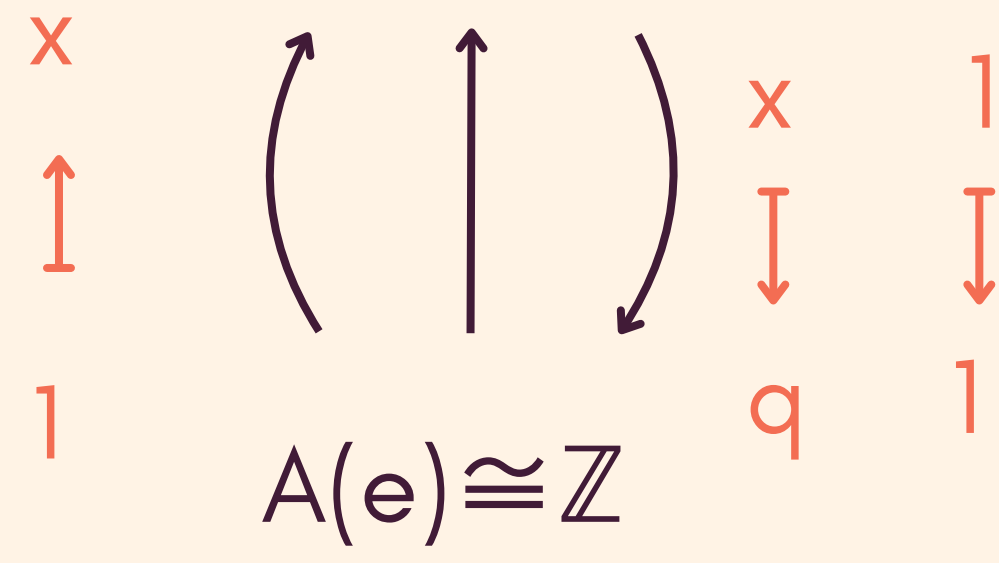


restriction: $\text{restriction} = \text{“id”}$

transfer: $\text{induction} = H \times_K (-)$

EX. $G = C_q$, q prime

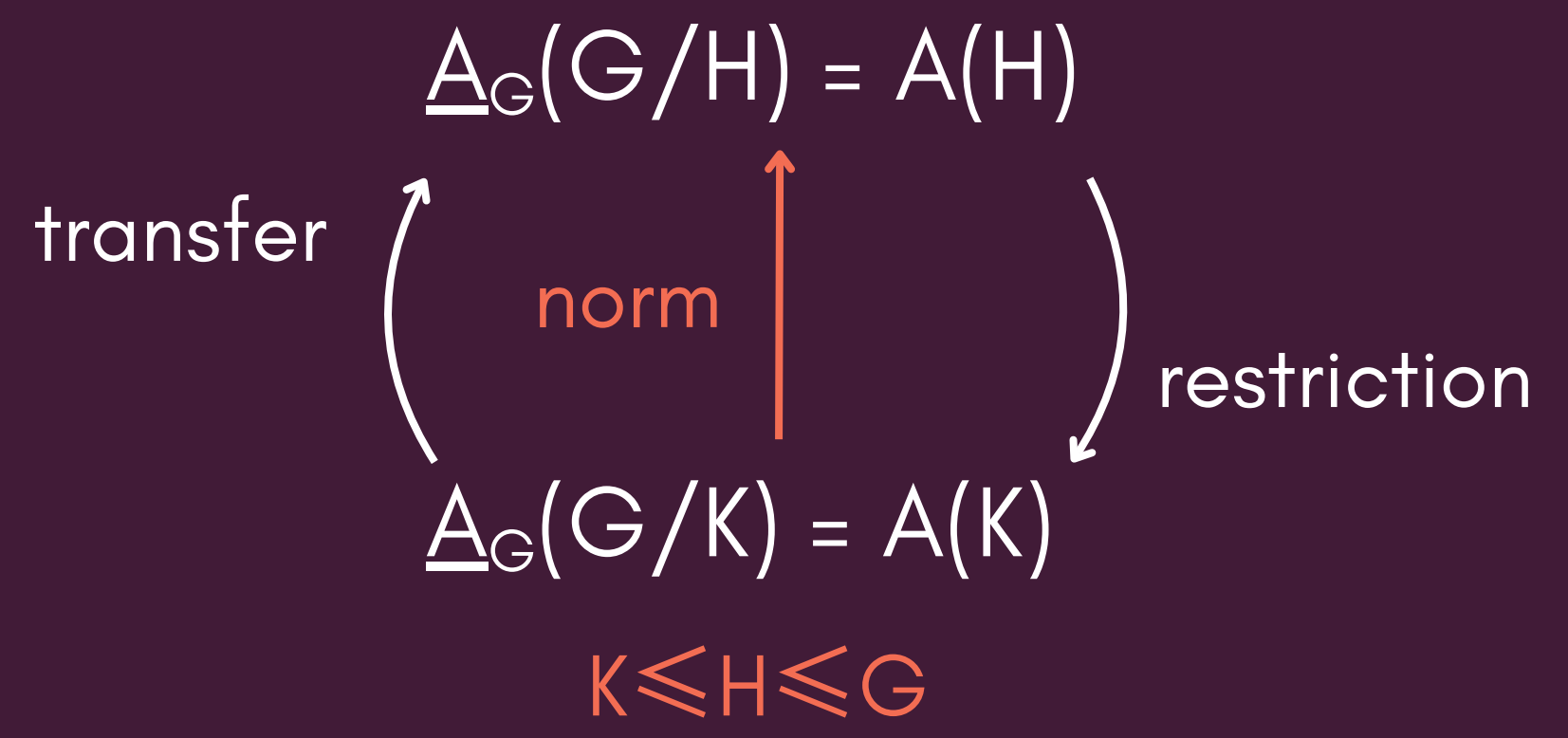
$$A(C_q) \cong \mathbb{Z}[x]/(x^2 - qx)$$



$$\text{norm}(a) = \frac{a^p - a}{p} x + a$$

WHAT IS $\underline{A}_G$ (REALLY)

“DEF.” The Burnside Tambara functor of G “is” $\underline{A}_G$:



restriction: restriction = “id”

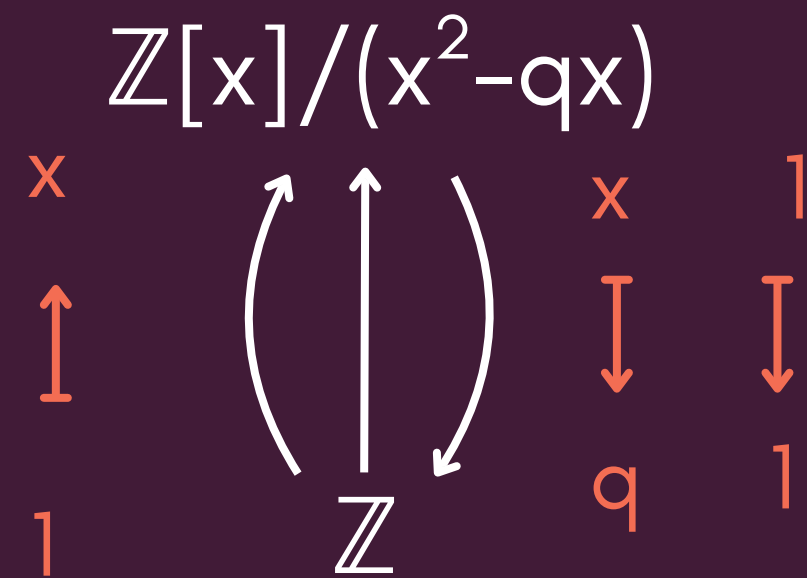
transfer: induction = $H \times_K (-)$

norm: coinduction = $\text{Map}_K(H, -)$

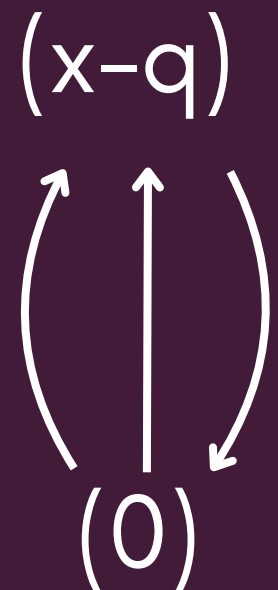
we can study it like a “ring”

TAMBARA IDEALS

EX. $G = C_q$, q prime



$$\text{norm}(a) = \frac{a^p - a}{p} x + a$$



prime ideal "over 0"

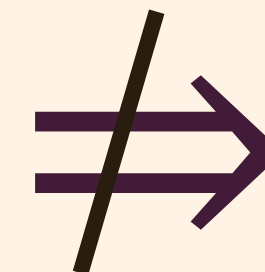
DEF. An ideal in a Tambara functor $\underline{M}$ is a collection of ideals $I(G/H) \subseteq \underline{M}(G/H)$ for all $H \leq G$ which are closed under the structure maps

DEF/PROP. An ideal $P = \{P(G/H)\}$ is prime iff for all $x \in P(G/H)$ and $y \in P(G/K)$,

all generalized products of x and y are in $P \implies x$ or y is in P



P is prime



$P(G/H)$ is prime

MAIN THEOREM (Nakaoka, C.-Ginnett, C.-CMQSVN)

For any finite group G ,

$$\text{Spec}(\underline{A}_G) = \{ P_{H,p} \mid p \text{ prime or } 0, H \leq G \}$$

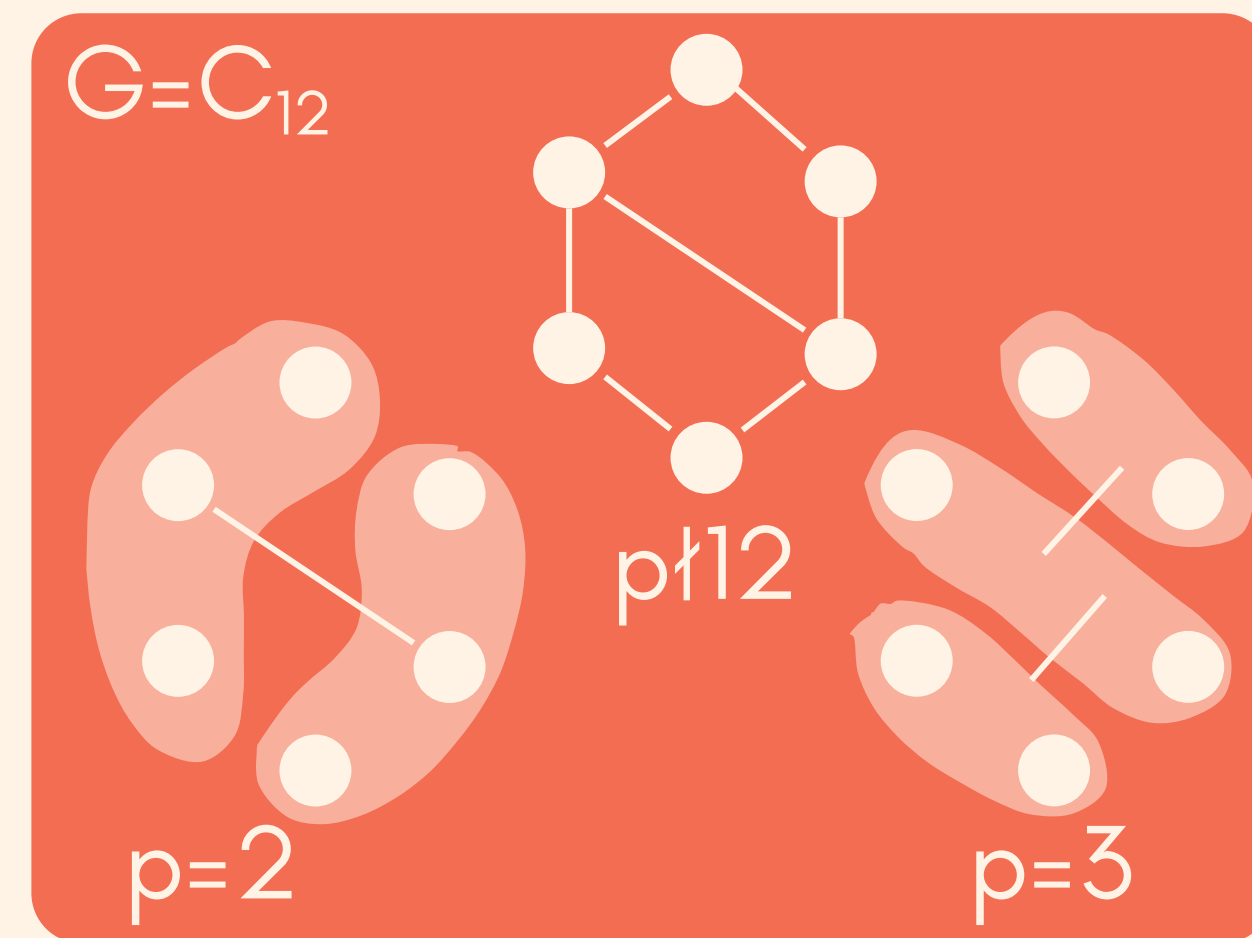
where $P_{H,p}(G/K) = \bigcap_{\substack{I \leq K \\ I \leq_G H}} \ker(\phi_{I,p}: A(K) \rightarrow \mathbb{Z}/p\mathbb{Z})$

and

- $P_{H,0} \subseteq P_{H,p}$ and $P_{H,p} \not\subseteq P_{H,0}$
- $P_{K,0} \subseteq P_{H,0}$ iff $H \leq_G K$
- $P_{K,p} \subseteq P_{H,q}$ iff $p=q$ and $O^p(H) \leq_G O^p(K)$
- $P_{K,0} \subseteq P_{H,p}$ iff $O^p(H) \leq_G K$

"p-residual subgroup"

e.g. $O^p(C_n) = C_m$
for $m = n/p^v$



COMPUTATIONAL TECHNIQUES

How do we usually classify prime ideals in R ?

- Study quotients R/I
- Study localizations $S^{-1}R$
- Study maps $R \rightarrow R'$

In equivariant algebra:

- Quotients are ok, but not always helpful
- Localizations are hard, and not always helpful
- Maps are ok, and...

THM. (CMQSVN) If $f: \underline{R} \rightarrow \underline{R}'$ is a levelwise integral extension of Tambara functors, then $f^* : \text{Spec}(\underline{R}') \rightarrow \text{Spec}(\underline{R})$ is surjective.

Proof idea:
like Dress,
construct a map
from $\underline{A}_G$ to “ghost”

THE GHOST CONSTRUCTION

“DEF.” The ghost of A_G “is”  :

$$\begin{array}{ccc}
 \text{transfer} \nearrow & \text{ghost}(G/H) = (\prod_{I \leq H} \mathbb{Z})^H & \searrow \text{restriction} \\
 & \text{norm} \uparrow & \\
 \text{ghost}(G/K) = (\prod_{J \leq K} \mathbb{Z})^K & &
 \end{array}$$

restriction of $(a_I)_{I \leq H} = (a_J)_{J \leq K}$

transfer of $(b_J)_{J \leq K} = (\sum_I b_I)_{J \leq K}$

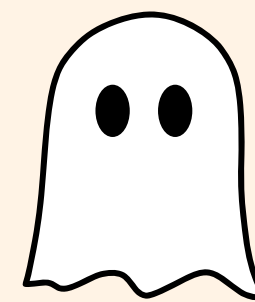
norm of $(b_J)_{J \leq K} = (\prod_I b_I)_{J \leq K}$

THM. (C.-CMQSVN) The ghost is a Tambara functor and Dress’ maps $A(H) \rightarrow \text{ghost}(G/H)$ assemble to a levelwise integral extension of Tambara functors

COR. The map $\text{Spec}(\text{ghost}) \rightarrow \text{Spec}(A_G)$ is surjective!

Q. What do prime ideals in  look like?

PRIME IDEALS IN



Observation: The ideals of  $(G/H) = (\prod_{I \leq H} \mathbb{Z})^H$ are specified by tuples $(n_{I,H})_{I \leq H}$ for some integers $n_{I,H}$

EX. Let F be a family of subgroups of G , and let $P(F,p) \subseteq$  be the ideal given by

$$n_{I,H} = \begin{cases} p & I \in F, \\ 1 & I \notin F. \end{cases}$$

PROP. For $H \leq G$ and $F_H = \{I \leq_G H\}$, the ideal $P_{H,p} := P(F_H,p)$ is prime.

THM. If P is prime, then $P = P_{H,p}$ for some H and p .

Key Lemma: If P is prime, then $n_{I,H} = n_{I,G} =: n_I$ is 0, 1, or p for all $I \leq H \leq G$

MORE COMPUTATIONS?

Is there a more general
“ghost” construction?
(see CMQSVN for $G=C_p$)

EQUIVARIANT TT- GEOMETRY???

(don't ask me)



THANK YOU!



OTHER INTERESTING EXAMPLES?

Besides $\text{Spec}(\underline{A}_G)$, we know
 $\text{Spec}(\underline{FP}(R)) \cong \text{Spec}(R^G)$
[CMQSVN]
...but not much else!

